

Customer Analytics

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Dataset: Online Retail II Period: Dec 2009 – Dec 2011 Customers: 5,878 Transactions: 779,425

Model: CatBoost Classifier ROC-AUC: 0.90 F1 Score: 0.80

Customer Segmentation

K-Means Clustering on RFM + Behavioral Features (K=4)

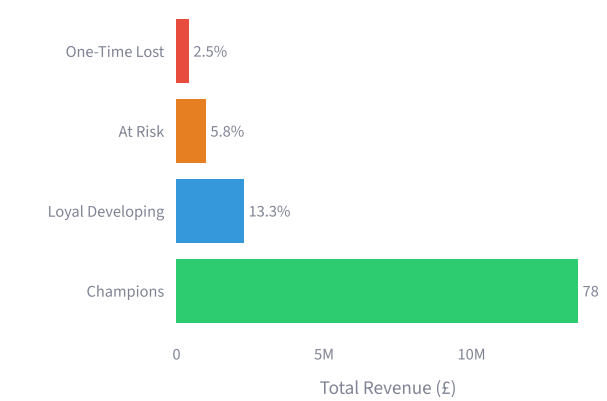
Select a segment to explore:

All Segments

Showing: 5,878 customers

Customers	Avg Monetary	Avg Frequency	Churn Rate
5,878	£868	3 orders	50.9%

Revenue Distribution by Segment



Frequency vs Monetary by Segment



Segment Profile Summary

	Persona	Customers	Median_Recency	Median_Frequency	Median_Monetary	Median_Tenure	Churn_Rate	Revenue_Share
0	At Risk	700	291.000000	2.000000	£736	82.000000	73.1%	5.8%
1	Champions	1520	25.000000	11.000000	£4216	619.500000	13.9%	78.4%
2	Loyal Developing	2031	71.000000	3.000000	£927	272.000000	43.5%	13.3%
3	One-Time Lost	1627	407.000000	1.000000	£217	0.000000	84.9%	2.5%

Business Recommendations

✔ Champions

Reward with VIP loyalty program and early product access. Low churn risk — focus on upselling and referral programs.

💡 Loyal Developing

Nurture with personalized email campaigns and cross-sell recommendations. Highest potential for conversion to Champions.

⚠ At Risk

Launch immediate win-back campaign with targeted discounts. High return rate signals dissatisfaction — prioritize service recovery.

🔔 One-Time Lost

Re-engagement campaign with strong incentive. Filter by Monetary value — focus budget on higher-value lost customers.